

BLM224 Elektronik

Hafta 7

Örnek Çözümleri

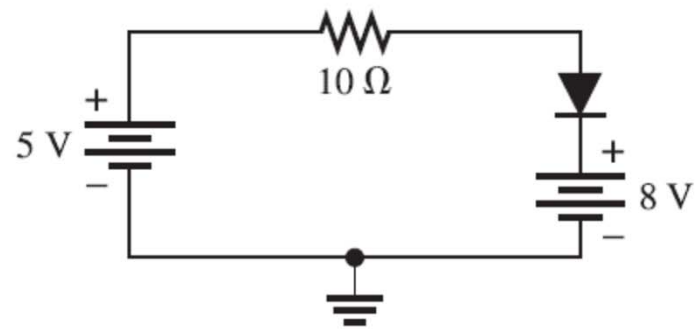
Karabük Üniversitesi
Bilgisayar Mühendisliği

Sınav İçin Hatırlatma

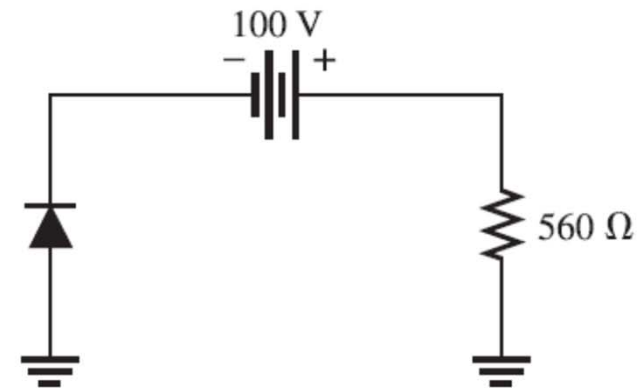
Yalnız 4 işlem yapan basit hesap makineleri sınavda kullanılabilir.

Beta değeri 50 ve üzerindeyse Kollektör ve Emitör akımlarını yaklaşık olarak birbirine eşit kabul ediniz.

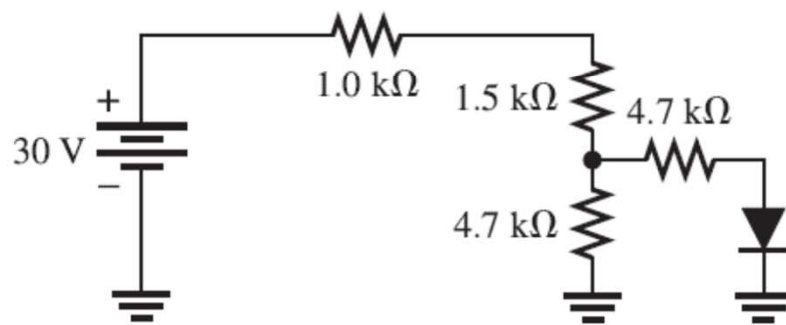
Örnek 1



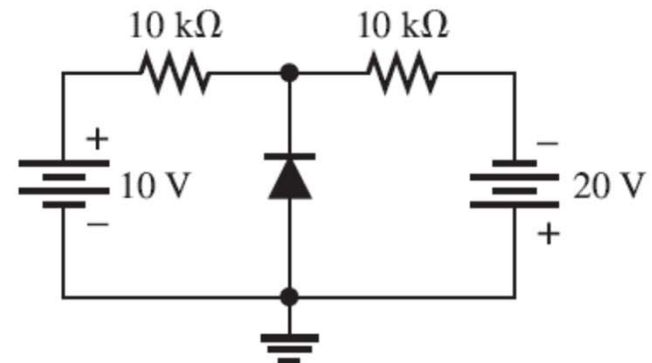
(a)



(b)

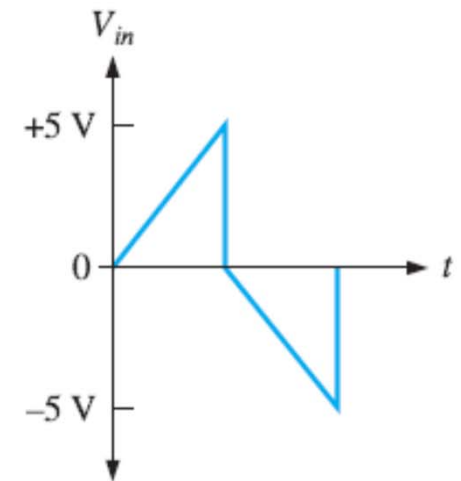
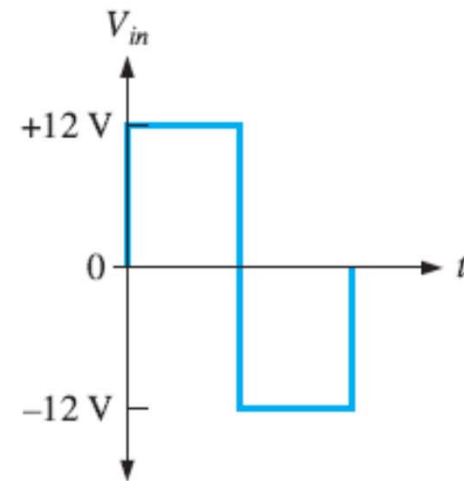
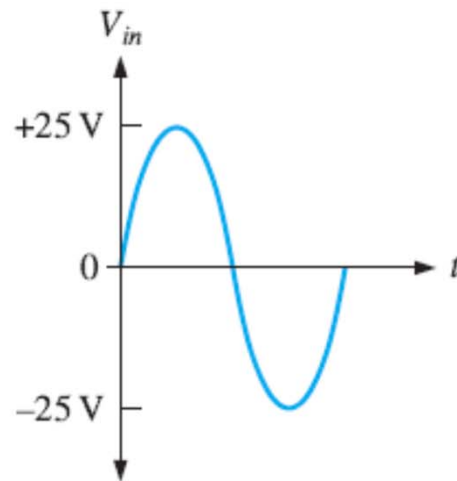
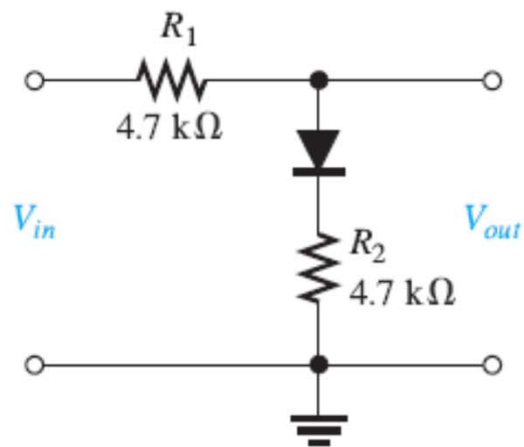


(c)

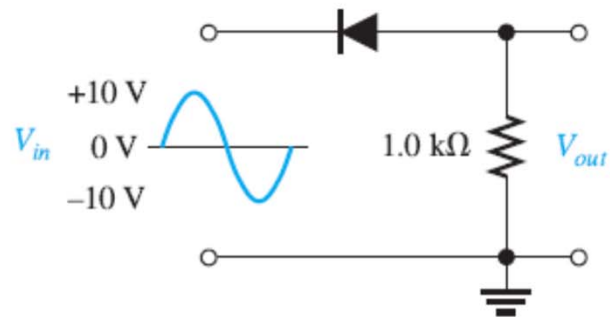


(d)

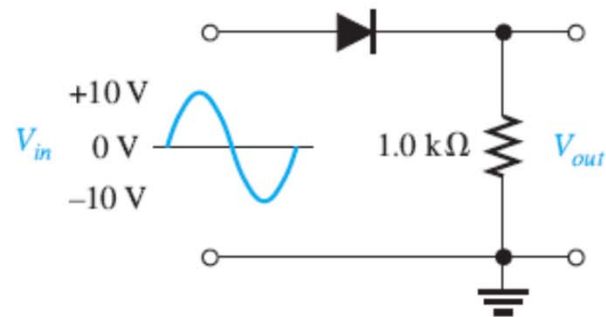
Örnek 2



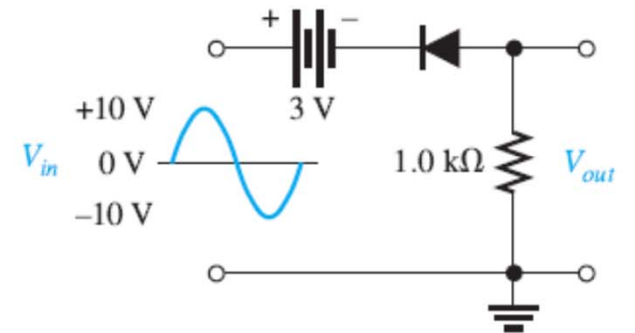
Örnek 3



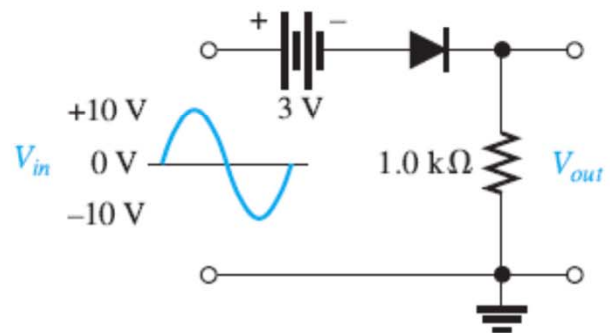
(a)



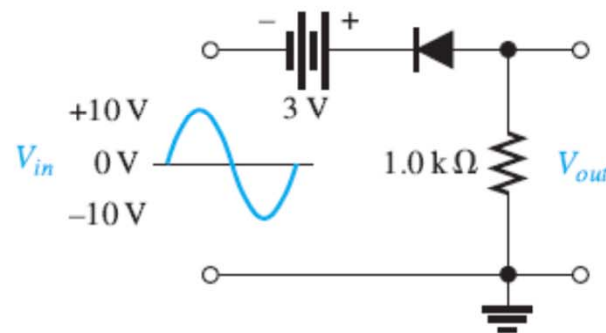
(b)



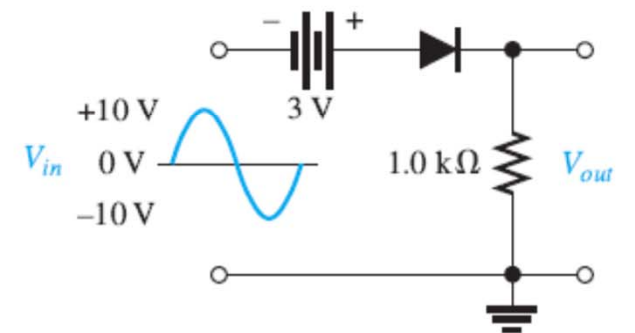
(c)



(d)

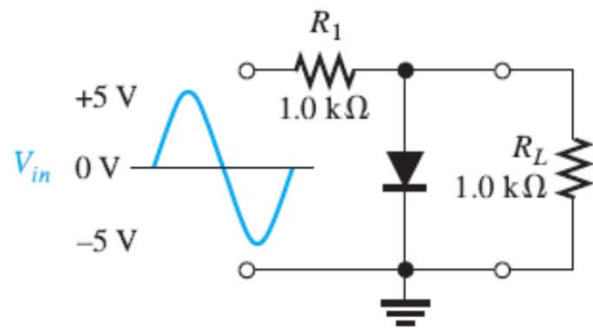


(e)

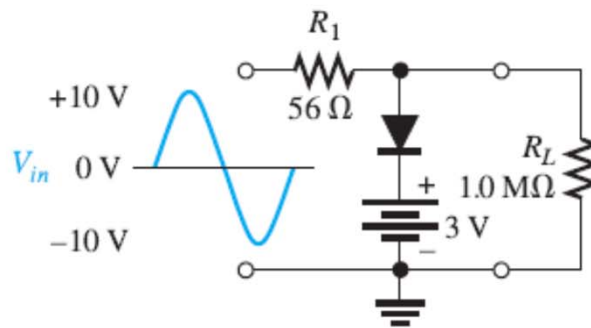


(f)

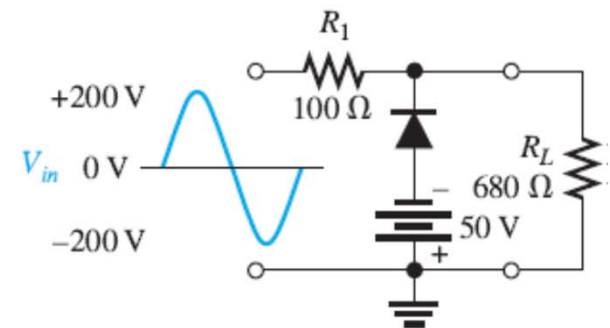
Örnek 4



(a)

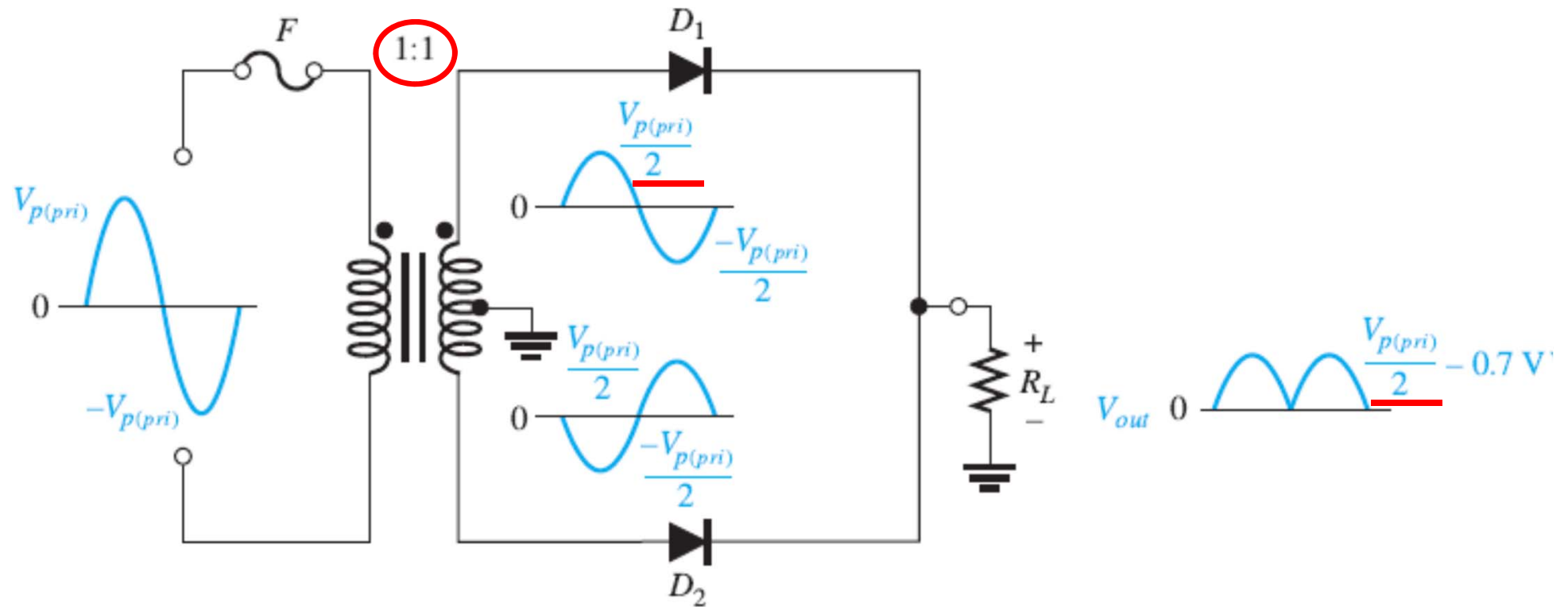


(b)

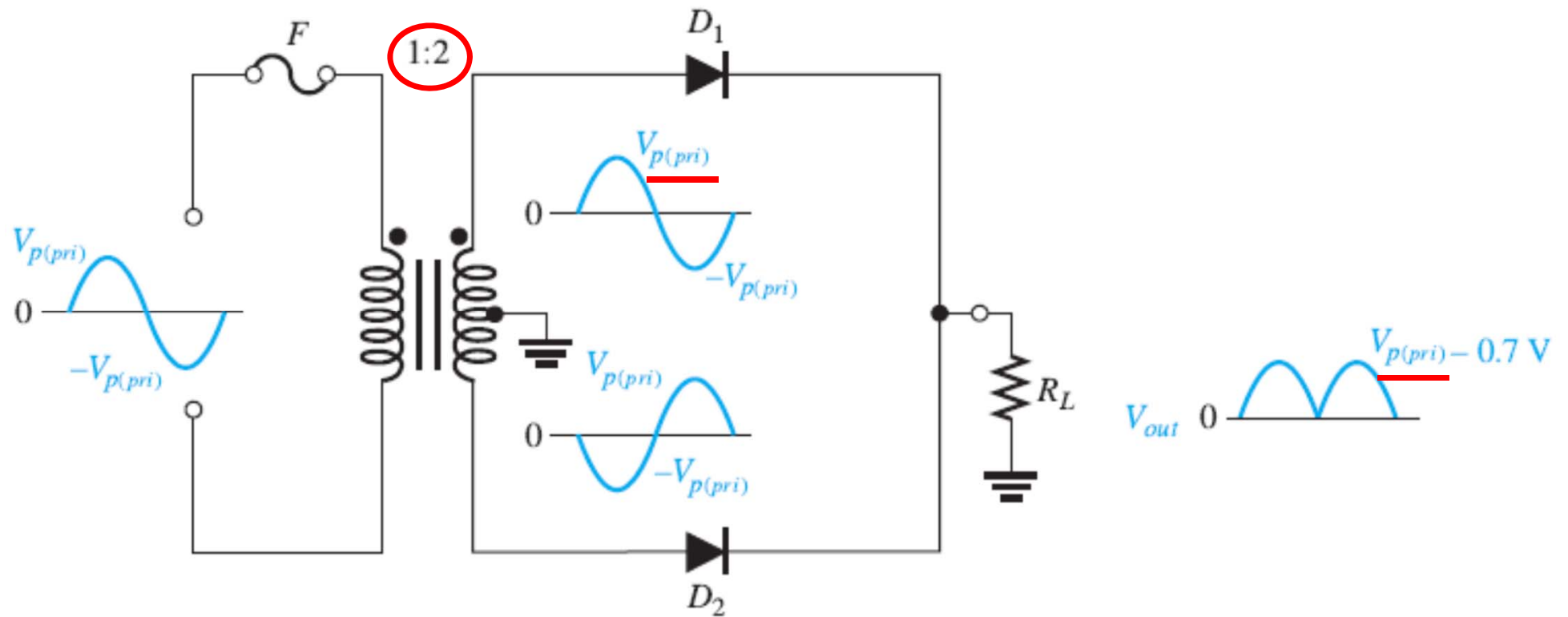


(c)

Hatırlatma

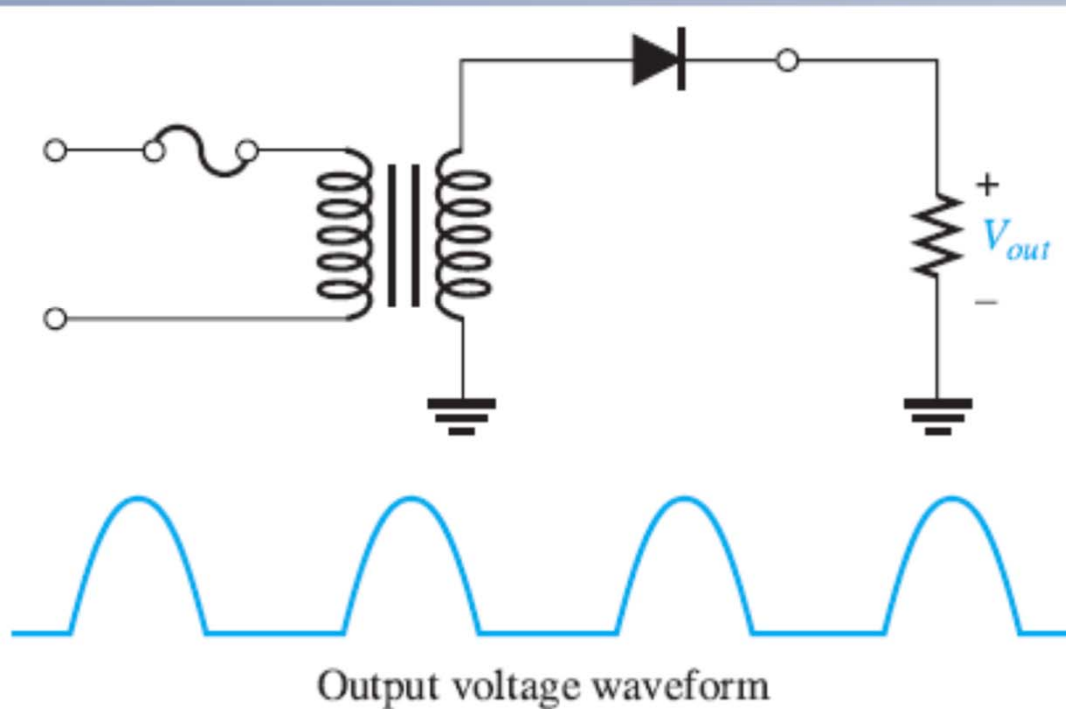


Hatırlatma



Hatırlatma

HALF-WAVE RECTIFIER



- Peak value of output:

$$V_{p(out)} = V_{p(sec)} - 0.7 \text{ V}$$

- Average value of output:

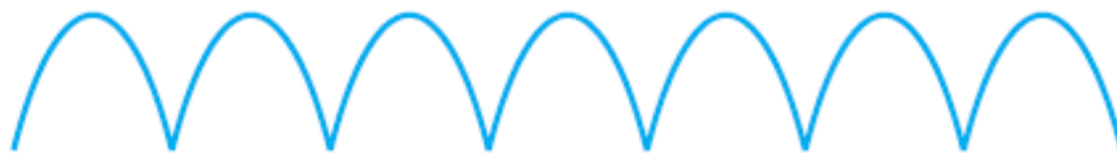
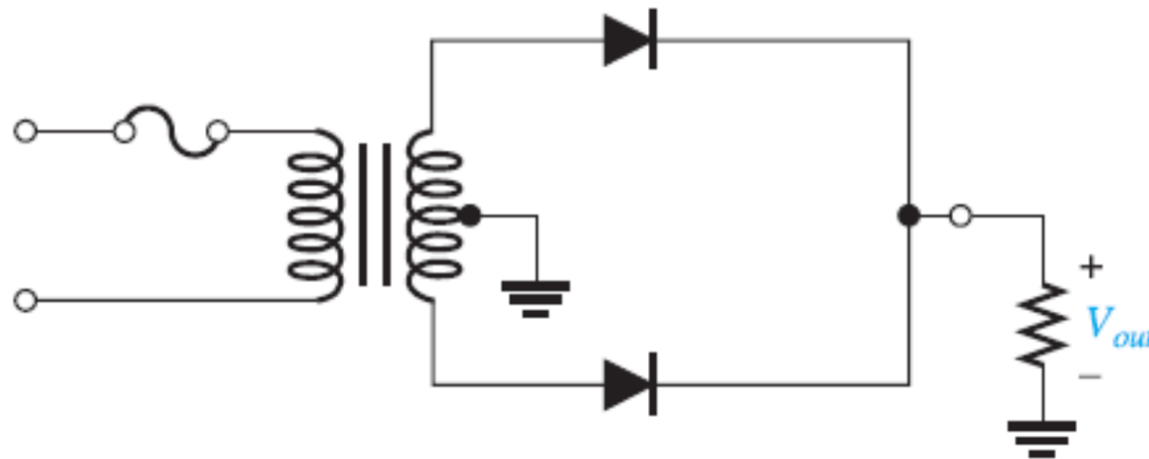
$$V_{AVG} = \frac{V_{p(out)}}{\pi}$$

- Diode peak inverse voltage:

$$PIV = V_{p(sec)}$$

Hatırlatma

CENTER-TAPPED FULL-WAVE RECTIFIER



Output voltage waveform

- Peak value of output:

$$V_{p(out)} = \frac{V_{p(sec)}}{2} - 0.7 \text{ V}$$

- Average value of output:

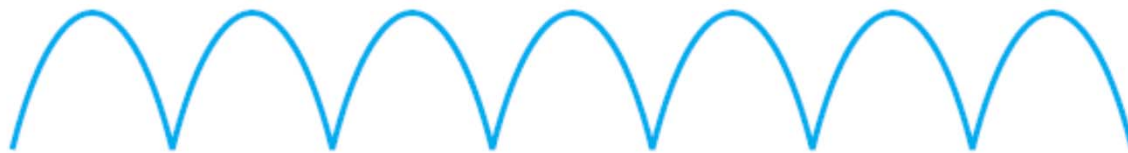
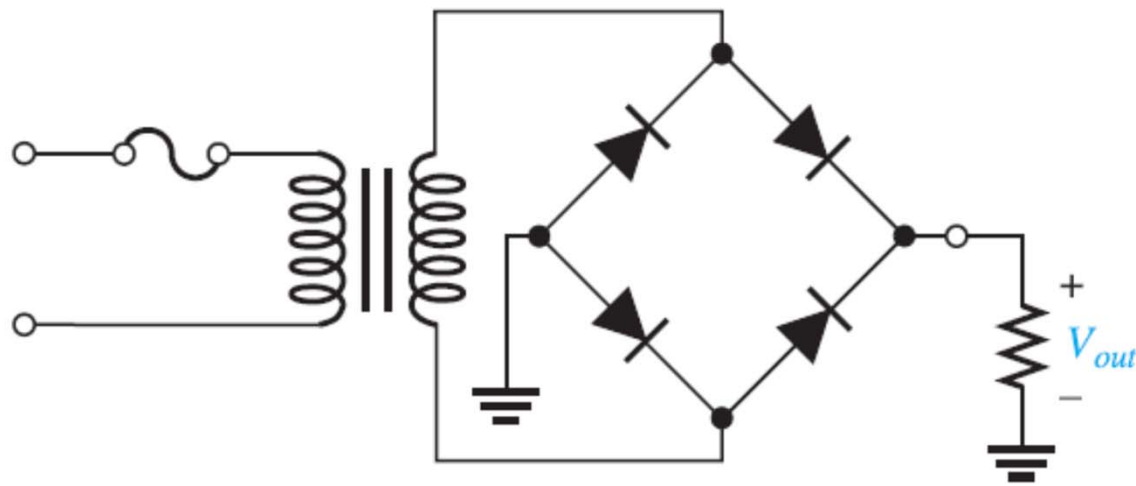
$$V_{AVG} = \frac{2V_{p(out)}}{\pi}$$

- Diode peak inverse voltage:

$$PIV = 2V_{p(out)} + 0.7 \text{ V}$$

Hatırlatma

BRIDGE FULL-WAVE RECTIFIER



Output voltage waveform

- Peak value of output:
$$V_{p(out)} = V_{p(sec)} - 1.4 \text{ V}$$
- Average value of output:
$$V_{AVG} = \frac{2V_{p(out)}}{\pi}$$
- Diode peak inverse voltage:
$$PIV = V_{p(out)} + 0.7 \text{ V}$$